# Practical No. 3

**Services, Notifications and Broadcast Receivers**

1. Develop an Android application implementing Started and Bound Services.
2. Develop an Android application implementing Notifications using Notification Manager.
3. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

**AIM:**To study and implement Programs on Services, notification and broadcast receivers Programs on Services, notification and broadcast receivers

- Started Service
- Notification from Service
- Broadcast Receiver (BOOT_COMPLETED)
- Buttons to Start/Stop Service
- Notification Channel (Android 8+)

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest package="com.example.servicedemo"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <!-- Permission for Boot Completed -->
    <uses-permission
android:name="android.permission.RECEIVE_BOOT_COMPLETED"/>
    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <!-- Service -->
        <service android:name=".MyService"/>
        <!-- Broadcast Receiver -->
        <receiver android:name=".BootReceiver"
            android:enabled="true"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.BOOT_COMPLETED"/>
            </intent-filter>
```

```
        </receiver>
        <!-- Main Activity -->
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <Button
        android:id="@+id/btnStart"
        android:text="Start Service"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnStop"
        android:text="Stop Service"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/btnNotify"
        android:text="Send Notification"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

**MainActivity.java**

```
package com.example.servicedemo;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    Button btnStart, btnStop, btnNotify;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnStart = findViewById(R.id.btnStart);
        btnStop = findViewById(R.id.btnStop);
        btnNotify = findViewById(R.id.btnNotify);
        // Start Service
        btnStart.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            startService(intent);
            Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show();
        })
```

```
        // Stop Service
        btnStop.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            stopService(intent);
            Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show();
        });
        // Send Notification via Service
        btnNotify.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            intent.setAction("SEND_NOTIFICATION");
            startService(intent);
        });
    }
}
```

**MyService.java**

```
package com.example.servicedemo;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.Service;
import android.content.Intent;
import android.os.Build;
import android.os.IBinder;
import androidx.core.app.NotificationCompat;
public class MyService extends Service {
   private static final String CHANNEL_ID = "ServiceChannel";
  @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
if (intent != null && "SEND_NOTIFICATION".equals(intent.getAction())) {
            sendNotification()  }
```

```
 return START_STICKY;
 }
private void sendNotification() {
 NotificationManager manager =
   (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
// Notification Channel (For Android 8+)
     if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        NotificationChannel channel = new NotificationChannel(
              CHANNEL_ID,
              "Service Channel",
              NotificationManager.IMPORTANCE_DEFAULT);
        manager.createNotificationChannel(channel);
     }
  NotificationCompat.Builder builder =
           new NotificationCompat.Builder(this, CHANNEL_ID)
                .setContentTitle("Service Notification")
                .setContentText("This is a Notification from Service")
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setAutoCancel(true);
 manager.notify(1, builder.build());
  }
@Override
  public IBinder onBind(Intent intent) {
     return null;
  }}
```

BootReceiver.java

```
package com.example.servicedemo;
import android.content.BroadcastReceiver;
import android.content.Context;
```

```
import android.content.Intent;
import android.widget.Toast;
public class BootReceiver extends BroadcastReceiver {
   @Override
    public void onReceive(Context context, Intent intent) {
if (Intent.ACTION_BOOT_COMPLETED.equals(intent.getAction())) {
     Toast.makeText(context,
           "Device Booted - Broadcast Received",
               Toast.LENGTH_LONG).show();
 // Optional: Start Service Automatically
         Intent serviceIntent = new Intent(context, MyService.class);
         context.startService(serviceIntent);
      }
   }}
```




